

The State of AI Agents

Executive Summary

AI agents are autonomous systems that observe, plan, and act. They are rapidly maturing and seeing real production use in 2025–2026.

Key Findings

Capability Growth

Models like GPT-4o and Claude 3.5 Sonnet can now reliably execute multi-step tasks with tool use.

- Long-context windows (up to 1M tokens) enable agents to hold entire codebases in context
- Native tool-calling APIs have replaced fragile prompt engineering patterns
- Structured JSON output makes agent pipelines more reliable

Failure Modes

Agents still struggle with ambiguous instructions and hallucination in long chains.

"The biggest risk in production agents is compounding errors — small mistakes at step 1 compound by step 10."

Analysis

The field is moving fast. Teams that build real evals now will outpace those who don't.

Sources

- <https://arxiv.org/abs/2401.01234> — Survey of LLM agents
- <https://www.anthropic.com/research/agent-failures> — Anthropic failure taxonomy
- <https://openai.com/research/function-calling> — OpenAI tool use docs with a very long URL that goes on forever and ever and should be truncated